

Lloyds BCB Data Collector

Dissertation Project - Complete README
Lloyds Banking Group · Commercial Banking Division · BCB Sector Intelligence

5 free APIs	Data sources
Zero	API keys needed
8 sectors	BCB sectors covered
~40 features	Cleaned dataset

1. Project Overview

This toolkit collects, processes, and merges publicly available data to identify commercial banking opportunity signals for Lloyds BCB (Business and Commercial Banking) target sectors. It supports the dissertation research question:

"Can external public data (company filings + media coverage) be used to predict business needs and inform commercial banking strategies?"

1.1 The 8 BCB Target Sectors

Lloyds specified the following sectors as priority targets for BCB engagement:

Sector	SIC Code Range	Key Characteristics
Manufacturing	10000 – 33999	Production, output signals
Public Sector, Education & Charities	84000 – 88999	Government-adjacent
Healthcare	86000 – 87999	NHS, private health
Technology, Legal & Professional	62000 – 74999	High growth potential
Agriculture	1000 – 3999	Seasonal, rural businesses
Real Estate	68000 – 68999	Commercial property
Wholesale & Retail	45000 – 47999	Consumer-facing
Fast Growth & Emerging	Cross-cutting	Identified by financial signals

2. Problem Statement

Lloyds Bank wants to grow its Business and Commercial Banking (BCB) division by finding UK companies that are likely to need banking products such as business loans, overdrafts, or corporate accounts. The challenge is that Lloyds does not have a ready-made list of "who needs what." Identifying these companies manually would require thousands of hours of research by relationship managers.

This project solves that problem by automatically collecting public data about 3.1 million UK companies and using machine learning to score each one on how likely it is to be a good banking prospect. Relationship managers can then focus on the top-ranked companies in their sector instead of guessing.

In plain English

We are building a system that looks at public information about every UK company (how old it is, whether it files accounts on time, how many directors it has, whether it already has loans, etc.) and predicts which ones are most likely to benefit from Lloyds commercial banking services.

2.1 Why Is This Hard?

The core difficulty is that we do not have ground truth data. We do not know which companies actually became Lloyds customers. Because of this, we cannot train a model the normal way (where you have a list of correct answers to learn from). Instead, we had to design proxy labels — our own definition of what a "good opportunity" looks like, based on observable public signals.

This is a known limitation and must be stated clearly in the dissertation. The model learns to predict our definition of opportunity, not confirmed Lloyds product uptake.

3. Challenges & How We Fixed Them

This section explains every major problem we encountered during the project and exactly how each one was resolved.

Challenge 1 — Broken Proxy Labels (Most Critical)

What went wrong:

When we first built the dataset, two of our three prediction targets were completely broken. The `label_growth_opportunity` and `label_risk_signal` columns showed 0 for every single company out of 3.1 million. A column that is always 0 has zero variance — a machine learning model cannot learn anything from it.

Why it happened:

The original label definitions depended on `avg_tone` — a news sentiment score from GDELT. The idea was: if a company has positive news coverage, mark it as a growth opportunity. If it has negative news coverage, mark it as a risk.

The problem is that GDELT tracks global news, and small private UK businesses almost never appear in any news at all. When we ran the GDELT queries, it returned zero articles for every single one of the ~350 companies we sampled. So `avg_tone` was empty (NULL) for 100% of the 3.1 million companies. Since the labels were built from `avg_tone`, they collapsed to 0 for everyone.

The result before the fix

`label_growth_opportunity`: 0% positive (0 out of 3,145,434 companies marked as 1) — completely useless.

`label_risk_signal`: 0% positive (0 out of 3,145,434 companies marked as 1) — completely useless.

`label_lending_need_proxy`: 13.1% positive — this one was fine because it did not depend on GDELT.

How we fixed it:

We completely rebuilt the two broken labels using Companies House data only — data that is available for every single company in the dataset. We stopped using news sentiment as the main ingredient and instead asked simple questions about each company using the structured data we already had.

Label	Old definition (broken)	New definition (fixed)
<code>label_growth_opportunity</code>	<code>avg_tone > 0</code> (positive news sentiment) → NULL for 100% of companies	Company age < 10 years AND is actively trading AND is not dormant AND has no overdue accounts
<code>label_risk_signal</code>	<code>avg_tone < -3</code> (negative news sentiment) → NULL for 100% of companies	Has ever filed accounts late OR conf. statement >180 days overdue OR has late filing history OR high director turnover

label_lending_need_proxy	Already working correctly	Has active mortgages AND is actively trading AND not flagged as risk
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News sentiment (avg_tone) is still kept in the dataset. But now it is used as a bonus adjustment only — if a company happens to have news data available, good sentiment slightly increases its label and bad sentiment slightly increases the risk label. It is no longer the main definition.

The result after the fix

label_growth_opportunity: 56.67% positive — 1,782,383 companies marked as 1. label_risk_signal: 6.58% positive — 207,014 companies marked as 1. label_lending_need_proxy: 12.90% positive — 405,783 companies marked as 1. All three labels now have meaningful, non-constant distributions ready for model training.

Challenge 2 — Massive Data Size

What the problem was:

The raw Companies House dataset contains every registered UK company — about 5 million rows. After filtering for our 8 target sectors we still had 3.1 million companies. The final merged dataset (final_model_dataset.csv) is 1.7 GB and the cleaned version (cleaned_companies_house_dataset.csv) is over 1 GB. These files cannot be loaded into memory all at once on a standard laptop.

How we fixed it:

We used chunked processing — instead of loading the entire file into memory at once, we read it in blocks of 200,000 rows at a time, processed each block, and wrote the results out. This kept memory usage manageable throughout. We also used more efficient data types (int8, float32 instead of int64, float64) to reduce the memory footprint of the final dataset by roughly half.

Additionally, because these files exceed GitHub's 100 MB file size limit, they are excluded from the repository via .gitignore. Only the code that generates them is tracked in git.

Challenge 3 — No Ground Truth Data

What the problem was:

In a normal machine learning project you have labelled examples — you know the right answer for each row and the model learns from those answers. Here, we do not have the right answers. We do not know which companies actually became Lloyds customers or which ones needed a loan. This is called the absence of ground truth.

How we handled it:

We designed proxy labels — our own engineered definition of what a growth opportunity, a risk signal, and a lending need looks like, based entirely on public observable signals from Companies House filings. These are reasonable approximations but they are not perfect. The dissertation must clearly state that these are proxy labels and that the model's predictions reflect our definition, not confirmed Lloyds outcomes.

Challenge 4 — Data Leakage Risk

What the problem is:

Because we built the labels ourselves from specific columns in the dataset, those same columns are still present as features. For example, label_growth_opportunity was built using company_age_years, is_active, is_dormant, and accounts_overdue_days. If you include those columns when training the model, the model will simply reconstruct the formula we used to create the label. It will score 99-100% accuracy, but that accuracy is completely fake — it has not learned anything real about businesses.

This is called data leakage

It is when information that was used to create the target label is also included as an input feature, allowing the model to "cheat" instead of actually learning.

How we handle it:

Before training, you must remove the columns that were used to build each label from the feature set. See Section 8 (Modelling Guide) for the exact list of columns to exclude for each label.

Challenge 5 — Sparse Enrichment Data

What the problem was:

The GDELT and OpenCorporates sources only sampled a few hundred companies from each sector. When that sparse data was merged onto the full 3.1 million company population, over 99% of rows had empty values for those enrichment columns. This means the dataset is almost entirely built from Companies House data in practice.

How we handled it:

Empty sentiment values (avg_tone) are filled with 0 during model training so they do not cause errors. The labels were rebuilt to not depend on these sparse columns. The limitation is documented so it can be discussed honestly in the dissertation.

4. Data Sources

All five sources are free to use. No credit card or paid subscription is required.

4.1 Companies House Bulk Download

What it is

The UK government keeps a register of every company registered in the UK. This is a monthly snapshot of all ~5 million companies — their name, industry code, filing dates, mortgage information, and status. It is the backbone of this entire project.

- Company name, registration number
- SIC codes (industry classification)
- Incorporation date and company status
- Accounts filing dates and overdue status
- Mortgage and charge information
- Account category (micro / small / full)

https://download.companieshouse.gov.uk/en_output.html

Free bulk download, no registration. File size: ~700MB ZIP.

4.2 Companies House Filing API

What it is

An API that gives detailed filing history and director appointment/resignation data per company. Free with optional key registration.

- Director appointment and resignation counts
- Late filing detection (days after deadline)
- Historical compliance pattern

Note: Works without a key but rate-limited to ~600 requests per 10 minutes.

4.3 GDELT

What it is

A global news tracking platform. We queried it for news articles about each sampled company to get sentiment scores. In practice, GDELT returned zero articles for all sampled companies — small private UK businesses do not appear in global news. GDELT data is kept in the dataset but does not contribute meaningfully to the labels.

4.4 OpenCorporates

What it is

A global company database useful for identifying cross-border subsidiaries and parent companies. Free tier limited to ~500 requests/day — only a small sample is enriched.

Source	What you get	API Key?	Cost	File
Companies House Bulk	All UK companies	No	Free	fetch_bulk.py
Companies House API	Filings, directors	Optional	Free	fetch_filings.py
GDELT	News sentiment	No	Free	fetch_gdelt.py
OpenCorporates	Cross-border data	No	Free	fetch_opencorporates.py

5. Project File Structure

```
lloyds_data_collector/  
  run_all.py          <- START HERE: runs all steps  
  requirements.txt    <- Python dependencies  
  Data_Cleaning.ipynb <- Step 7: clean and prepare final dataset  
  
  companies_house/  
    fetch_bulk.py      <- Step 1: download bulk data  
    filter_sectors.py  <- Step 2: filter for BCB sectors  
    fetch_filings.py   <- Step 3: director & filing data  
  
  gdelt/  
    fetch_gdelt.py     <- Step 4: news sentiment  
  
  news/  
    fetch_opencorporates.py <- Step 5: cross-border data  
  
  utils/  
    sic_codes.py       <- SIC code to sector mapping  
    merge_all.py       <- Step 6: join all sources & create labels  
  
  output/  
    companies_bcb_filtered.csv  
    filings_and_directors.csv  
    gdelt_sentiment.csv  
    gdelt_sector_trends.csv  
    opencorporates_enriched.csv  
    final_model_dataset.csv <- 3.1M rows x 110 cols (too large for GitHub)  
    cleaned_companies_house_dataset.csv <- FINAL: 3.1M rows x 40 cols, ready for model
```

6. Setup & Installation

6.1 Requirements

- Python 3.8 or higher

- Internet connection (for downloads and API calls)
- ~2GB free disk space (for bulk download)
- 15–30 minutes run time depending on connection speed

6.2 Install Dependencies

```
pip install -r requirements.txt
```

6.3 Run Everything

```
python run_all.py
```

This runs all 6 steps in order. If a step fails it will ask whether to continue.

7. Dataset Columns — Explained in Simple Words

The cleaned dataset has 40 columns across 3.1 million UK companies. Here is what each column means in plain language.

7.1 Who the Company Is (Identity — remove before training)

Column	Simple meaning
CompanyName	The company's registered name
CompanyNumber	The unique 8-digit ID every UK company gets when registered
bcb_sector	Which of the 8 Lloyds target industries this company belongs to
SICCode.SicText_1	The specific industry description from Companies House

7.2 What We Know About the Company (Features — model learns from these)

Column	Simple meaning
company_age_years	How many years since the company was registered. Newer companies may need more banking support.
is_active	Is the company currently trading? 1 = yes, 0 = no. Only active companies are real prospects.
is_dormant	Is it a shell/inactive company with no real business activity? 1 = yes (skip it).
is_micro	Very small company — annual turnover under £632k.
is_small	Small company — annual turnover under £10.2 million. These are Lloyds' core BCB targets.
is_full_accounts	Large enough to file full accounts — a 'bankable' sized business. 1 = yes.
has_changed_name	Has this company ever traded under a different name? Could signal a rebrand or restructure.
accounts_overdue_days	How many days past the deadline are their annual accounts? 0 = healthy. Over 100 = concerning.
accounts_ever_late	Have they ever filed their accounts late at least once? 1 = yes.
conf_stmt_overdue_days	How many days late is their annual confirmation statement? (Similar to a census for companies.)
has_late_filing_history	Do they have a repeated pattern of filing things late? 1 = yes.
num_mortgages_outstanding	How many active business loans or charges are registered against this company.
has_active_mortgages	Does the company have any active loans registered? 1 = yes.

director_turnover_signal	Did an unusually high number of directors leave recently? 1 = yes. Can signal instability.
director_resignations_1yr	Exact count of directors who resigned in the past 12 months.
avg_tone	GDELT news sentiment score (-10 to +10). Positive = good press, negative = bad press. NULL for almost all companies.
pct_negative	What percentage of news articles about this company had a strongly negative tone.
sector_avg_tone	How is this company's whole industry sector being talked about in the news on average.
sector_volume_trend	Is news coverage of this sector growing or shrinking over time?

7.3 What We Are Predicting (Labels — the targets)

Important — these are proxy labels, not real answers

We invented these labels from public data because we do not have real Lloyds customer data. The model learns to predict our definition of opportunity, not confirmed banking outcomes. This must be stated clearly in the dissertation.

Label	Simple meaning	Distribution after fix
label_growth_opportunity	Is this a young, active, healthy company that looks like it is growing and could benefit from Lloyds banking services? This is the PRIMARY prediction target.	56.67% = yes (1.78M companies)
label_risk_signal	Does this company show warning signs — late filings, overdue statements, high director turnover? Use as a secondary model.	6.58% = yes (207K companies)
label_lending_need_proxy	Does this company already have active loans and is it actively trading? It may need more lending. Use with caution — hardest to defend as a label.	12.90% = yes (406K companies)

8. Modelling Guide

8.1 What the Model Does

The model reads all the feature columns for each company and learns to predict label_growth_opportunity (1 = good prospect, 0 = not). Once trained, it outputs a probability score between 0 and 1 for every company. Relationship managers can then sort by this score to find the best prospects in each sector.

8.2 Data Leakage Warning — Read Before Training

Remove label-defining columns before training

The labels were built from specific columns. Including those same columns as model inputs will let the model trivially reconstruct the label formula, giving fake near-perfect accuracy. You MUST exclude them. For label_growth_opportunity: exclude company_age_years, is_active, is_dormant, accounts_overdue_days. For label_risk_signal: exclude accounts_ever_late, conf_stmt_overdue_days, has_late_filing_history, director_turnover_signal. For label_lending_need_proxy: exclude has_active_mortgages and label_risk_signal.

8.3 Train/Test Split

```
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.read_csv('cleaned_companies_house_dataset.csv')

# Remove label-defining columns to prevent data leakage
FEATURES = [
```

```

    'is_micro', 'is_small', 'is_full_accounts', 'has_changed_name',
    'num_mortgages_outstanding', 'has_active_mortgages',
    'director_resignations_1yr',
    'avg_tone', 'pct_negative',
    'sector_avg_tone', 'sector_volume_trend'
]
TARGET = 'label_growth_opportunity'

X = df[FEATURES].fillna(0)
y = df[TARGET]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

```

8.4 Train the Model (XGBoost)

```

from xgboost import XGBClassifier
from sklearn.metrics import classification_report

model = XGBClassifier(random_state=42, eval_metric='logloss')
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))

```

8.5 Generate the Final Ranked Output

After training, score all 3.1 million companies and sort by predicted probability. The top-ranked companies in each sector are the ones Lloyds should approach first.

```

df_out = df[['CompanyNumber', 'CompanyName', 'bcb_sector'] + FEATURES].copy()
df_out['predicted_label'] = model.predict(X)
df_out['predicted_prob'] = model.predict_proba(X)[:, 1]
df_out = df_out.sort_values('predicted_prob', ascending=False)
df_out.to_csv('output/predictions.csv', index=False)

```

The predicted_prob column (0.0 to 1.0) is the key output. Higher = stronger prospect. Sort by this column to get a ranked list for each sector.

9. Dissertation Methodology Notes

9.1 How to Describe the Proxy Labels

Suggested dissertation wording

"In the absence of ground-truth commercial banking uptake data, binary proxy labels were constructed from publicly observable company characteristics — including filing compliance, company age, trading status, mortgage activity, and director turnover — as surrogate measures of growth opportunity, risk, and lending need. News sentiment data from GDELT was incorporated as a supplementary signal where available; however, coverage of small private UK companies was found to be negligible in practice, and all three labels are therefore primarily derived from structured Companies House data."

9.2 Key Limitations to State in the Dissertation

- Proxy labels are not ground truth — they approximate likely need, not confirmed product uptake.
- GDELT returned zero news articles for all sampled companies. Small private UK firms do not appear in global news. The avg_tone column is NULL for ~100% of the population and is filled with 0 during model training.
- Companies House bulk data is a monthly snapshot — some information may be slightly outdated.

- OpenCorporates free tier has a ~500 request/day limit — only a small sample of companies is enriched with cross-border data.
- The Fast Growth sector is identified by financial signals rather than a specific SIC code range — this approximation should be acknowledged.
- Label-defining features must be excluded from the model to avoid data leakage (see Section 8.2).

9.3 Key Strengths to Highlight

- Entirely replicable using free, public data — no proprietary or paid sources used.
- Covers all 8 Lloyds BCB target sectors systematically across 3.1 million companies.
- Labels were rebuilt from population-wide signals after the GDELT coverage gap was identified, ensuring all three labels have non-degenerate distributions.
- Combines structured (Companies House) and unstructured (GDELT news) data types as specified.
- Produces an actionable ranked output that relationship managers can use directly.
- Feature importance from XGBoost provides explainability — the model can justify each prediction.

10. Troubleshooting

Problem	Fix
Bulk download URL not found	Go to download.companieshouse.gov.uk , copy the latest URL manually into <code>fetch_bulk.py</code>
GDELT returns empty results	Expected — small UK companies rarely appear in global news. Labels no longer depend on GDELT. NULL <code>avg_tone</code> fills to 0 in training.
All label values are constant (0 or 1)	Run <code>merge_all.py</code> again — the updated label logic builds labels from Companies House columns, not GDELT.
Model accuracy is suspiciously near 100%	Data leakage — you have included label-defining columns in your feature set. See Section 8.2 for the correct exclusion list.
OpenCorporates 429 error	Rate limit hit. Script pauses 60 seconds automatically. Reduce <code>SAMPLE_SIZE</code> if still failing.
Memory error when loading CSV	The file is 1.7 GB. Close other applications. Use chunked reading: <code>pd.read_csv(..., chunksize=200000)</code> .
XGBoost import error	Run: <code>pip install xgboost</code>
<code>final_model_dataset.csv</code> missing columns	Run <code>merge_all.py</code> only AFTER all other scripts (<code>fetch_bulk</code> , <code>filter_sectors</code> , <code>fetch_filings</code> , <code>fetch_gdelt</code> , <code>fetch_opencorporates</code>) have completed.